

A Factorization Method Approach to the Biharmonic Transmission Problem in Absorbing Media

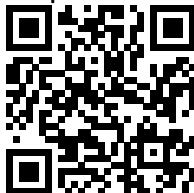
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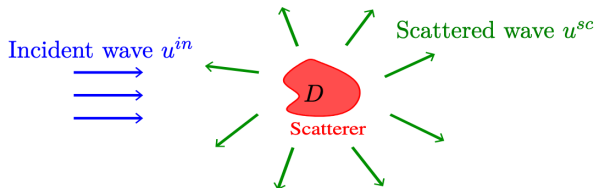
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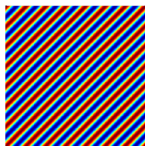
Paper on arXiv:



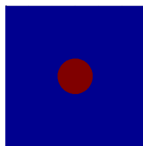
Direct & Inverse Scattering



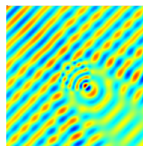
- 1 **Direct Problem:** Given incident wave u^{in} and scatterer D , find the scattered wave u^{sc} .
- 2 **Inverse Problem:** Given the (far-field) of scattered wave u^{sc} , find the scatterer D .



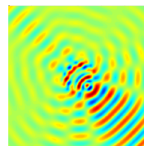
u^{in}



scatterer



u



u^{sc}

Kirchhoff–Love Plates and Flexural Waves

- ▶ A **Kirchhoff–Love thin plate** models out-of-plane bending of a homogeneous elastic plate of small thickness h .
- ▶ Transverse displacement $u(x)$ satisfies the **biharmonic wave equation**

$$\mathcal{D} \Delta^2 u - \rho h \omega^2 u = 0, \quad \mathcal{D} = \frac{Eh^3}{12(1 - \mu^2)}.$$

- E : Young's modulus– measure of stiffness, resistance
- ρh : mass per unit area
- $\mu \in (0, 1/2]$: Poisson's ratio

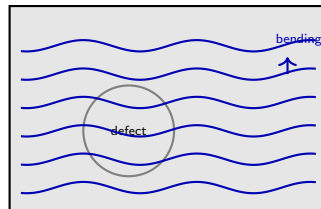
- ▶ Flexural waves obey

$$\kappa^4 = \frac{\rho h \omega^2}{\mathcal{D}}, \quad u = u^{\text{in}} + u^{\text{sc}}.$$

- ▶ A localized defect/inclusion modifies the rigidity:

$$\mathcal{D}(x) \neq \mathcal{D} \quad \text{in defect.}$$

- ▶ Leads to a **transmission problem for bending waves**.



Setting: Biharmonic Transmission Problem

- Out-of-plane displacement $u = u^{\text{in}} + u^{\text{sc}}$ satisfies

$$(\Delta^2 - \kappa^4 n(x)) u(x) = 0 \quad \text{in } \mathbb{R}^2, \quad n \in L^\infty(\mathbb{R}^2), \quad \text{supp}(n - 1) = \overline{D}.$$

- Incident field:

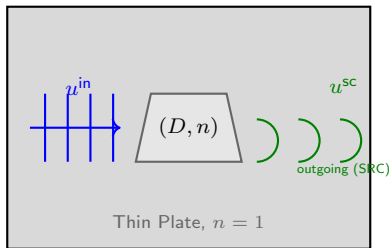
$$u^{\text{in}}(x, d; \kappa) = e^{i\kappa x \cdot d}, \quad d \in \mathbb{S}^1.$$

- Transmission conditions on ∂D :

$$[[u]] = [[\partial_\nu u]] = [[\Delta u]] = [[\partial_\nu \Delta u]] = 0.$$

- Sommerfeld radiation conditions for the biharmonic operator:

$$\lim_{r \rightarrow \infty} r^{1/2} (\partial_r u^{\text{sc}} - i\kappa u^{\text{sc}}) = 0, \quad \lim_{r \rightarrow \infty} r^{1/2} (\partial_r (\Delta u^{\text{sc}}) - i\kappa \Delta u^{\text{sc}}) = 0.$$



Absorbing Scatterer Condition and Well-Posedness

Direct Problem: Given the incident field u^i , find the scattered field u^s for a known penetrable medium (n, D) :

$$(\Delta^2 - \kappa^4 n(x)) u^{\text{sc}} = \kappa^4 (n(x) - 1) u^{\text{in}} \quad \text{in } \mathbb{R}^2, \quad u^{\text{sc}} \& \Delta u^{\text{sc}} \text{ satisfies SRC.}$$

- ▶ Well-posedness (Fredholm of index zero) is established via a variational approach.
- ▶ **Uniqueness:** In the biharmonic case, requires the **absorbing scatterer condition**

$$\Im(n(x)) \geq \alpha > 0 \quad \text{a.e. } x \in D$$

which is *not* necessary in the acoustic case.

Theorem (Ceja-Ayala, R., Harris, I. & Sánchez-Vizuet, T., 2025)

For a penetrable medium (n, D) satisfying $\Im(n) \geq \alpha > 0$ a.e. in D and a given incident field u^{in} :

- ▶ The direct scattering problem has a unique solution u^{sc} .
- ▶ The solution operator is Fredholm of index zero.
- ▶ The scattered field u^{sc} satisfies the estimate

$$\|u^{\text{sc}}\|_{H^2(D)} \leq C \|u^{\text{in}}\|_{L^2(D)}$$

Lippmann–Schwinger Type Representation

Goal: Express the **unique** scattered field solution u^{sc} in terms of volume integral

Idea: Apply the Green's theorem for x in both the *interior* and *exterior* of the scatterer D . (Note χ_D is characteristic function over D)

Lippmann–Schwinger Type Representation

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Interior ($x \in D$)

$$u^{\text{sc}}(x)\chi_D = \int_D G(x, z; \kappa) [\kappa^4 u^{\text{sc}}(z) - \Delta^2 u^{\text{sc}}(z)] dz + \text{boundary terms on } \partial D.$$

Exterior ($x \notin D$)

$$u^{\text{sc}}(x)(1 - \chi_D) = \text{boundary terms on } \partial D \cup \partial B_R.$$

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Exterior ($x \notin D$)

$$u^{\text{sc}}(x)(1 - \chi_D) = \text{boundary terms on } \partial D \cup \partial B_R.$$

Combine both: adding the interior and exterior representations gives

Lippmann-Schwinger (L-S) Type Equation

$$u^{\text{sc}}(x, d) = \kappa^4 \int_D (n(z) - 1) G(x, z; \kappa) [u^{\text{in}}(z, d) + u^{\text{sc}}(z, d)] dz.$$

Interpretation:

► $G(x, z; \kappa)$ is the Green's function of biharmonic operator $(\Delta^2 - \kappa^4)$.

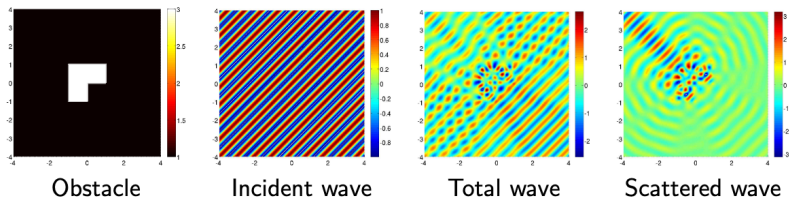
The Inverse Problem

Far-field intuition: Far from the scatterer, the scattered wave behaves like an outgoing cylindrical wave. Its amplitude and phase in each observation direction define the **far-field pattern**.

$$u^{\text{sc}}(x, d) = \frac{e^{i\pi/4}}{\sqrt{8\pi\kappa}} \frac{e^{i\kappa|x|}}{\sqrt{|x|}} u^\infty(\hat{x}, d; \kappa) + O(|x|^{-3/2}), \quad |x| \rightarrow \infty, \quad \hat{x} = \frac{x}{|x|}.$$

- $u^\infty(\hat{x}, d; \kappa)$: measured far-field pattern in direction of observation $\hat{x}$ for incident waves $d \in \mathbb{S}^1$
- **Inverse problem:** Recover the ‘**absorbing**’ penetrable scatterer characterized by the pair $(n, D = \text{supp}(n - 1))$ — its shape, size, and location — given the full far-field data

$$\{u^\infty(\hat{x}, d; \kappa) : \hat{x} \in \mathbb{S}^1, d \in \mathbb{S}^1\}.$$



Adapted from: A. Lechleiter, *Making the Invisible Visible: Imaging Techniques for Inverse Problems* (2009)

We now define the (compact!) far-field operator as $F : L^2(\mathbb{S}^1) \rightarrow L^2(\mathbb{S}^1)$

$$(Fg)(\hat{x}) = \int_{\mathbb{S}^1} u^\infty(\hat{x}, d; \kappa) g(d) ds(d).$$

Fg is far field corresponding to the incident field (Herglotz wave function)

$$v_g(x) = \int_{\mathbb{S}^1} e^{i\kappa x \cdot d} g(d) ds(d)$$

Note: v_g refers to superposition of plane waves.

Inverse Problem

The **inverse problem** also reads: given F find $D = \text{supp}(n - 1)!$

How to reconstruct D from F ? Nonlinear optimization, local linearization, domain decomposition... **time-consuming!, Requires forward solver!**

Qualitative/Sampling Methods

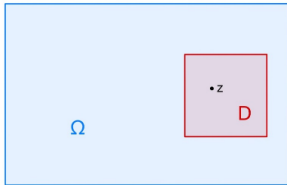
Class of Reconstruction Methods

Qualitative Methods: Construct a binary criterium that decides whether a grid point z belongs to the support of the region in a computational simple manner.

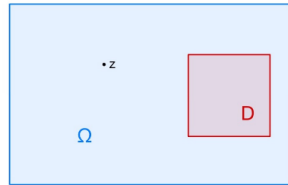
- **Advantages:** Requires **little a priori** information on the scatterer. Very fast, no forward solver needed, fully direct method.
- **Disadvantages:** requires a lot of data, can only recover qualitative information.

e.g., linear sampling method (Colton-Kirsch, 1996), factorization method (Kirsch, 1998), probe method (Ikehata, 1999), singular source method (Potthast, 2000), etc.

$$W(z) > 0$$



$$W(z) = 0$$



The Range Test Property

Define the **Herglotz wave operator**

$$(Hg)(x) = v_g(x), \quad x \in D, \quad L^2(\mathbb{S}^1) \rightarrow L^2(D) \quad \text{bounded}$$

and its adjoint

$$(H^* \varphi)(\hat{x}) = \int_D e^{-i\kappa \hat{x} \cdot y} \varphi(y) dy, \quad \hat{x} \in \mathbb{S}^1, \quad L^2(D) \rightarrow L^2(\mathbb{S}^1) \quad \text{bounded}$$

For a sampling point $z \in \mathbb{R}^2$, consider the **range test equation**

$$H^* \varphi = \phi_z, \quad \text{where } \phi_z(\hat{x}) = e^{-i\kappa \hat{x} \cdot z}.$$

Note: ϕ_z is a **test function** and far-field pattern for the fundamental solution of the Helmholtz equation

Lemma (Range Test Property), cf. Theorem 4.6 (Kirsch-Grinberg book)

Let $D = \text{supp}(n - 1)$ be a bounded open set with connected complement. Then,

$$\phi_z \in \text{Range}(H^*) \iff z \in D.$$

Key. Proof via a unique continuation argument.

H^* is a **bounded operator** depending on the (unknown) domain D .

Key Question

Can we relate $\text{Range}(H^*)$, which depends on the unknown D , to the **known far-field operator** F ?

Preliminaries for Factorization

Scattered field formulation. Let w solve the inhomogeneous biharmonic scattering problem

$$(\Delta^2 - \kappa^4)w = \kappa^4(n-1)(w+f) \quad \text{in } \mathbb{R}^2,$$

with Sommerfeld radiation conditions for w and Δw . Then w admits the Lippmann-Schwinger representation

$$w(x) = \kappa^4 \int_D (n(y) - 1) G(x, y) [w(y) + f(y)] dy.$$

Biharmonic Green's function. The outgoing fundamental solution of

$$(\Delta^2 - \kappa^4)G(\cdot, y) = \delta_y$$

is

$$G(x, y) = \frac{i}{8\kappa^2} \left[H_0^{(1)}(\kappa|x-y|) - H_0^{(1)}(i\kappa|x-y|) \right].$$

Far-field asymptotics as $|x| \rightarrow \infty$. Let $r = |x|$, $\hat{x} = x/r$. Then

$$G(x, y) \sim \frac{e^{i\kappa r}}{\sqrt{r}} \underbrace{\frac{\gamma}{2\kappa^2} e^{-i\kappa \hat{x} \cdot y}}_{\text{far-field term}}, \quad \gamma = \frac{e^{i\pi/4}}{\sqrt{8\pi\kappa}}.$$

Factorization of F via Lippmann–Schwinger

Starting point:

$$w(x) = \kappa^4 \int_D (n(y) - 1) G(x, y) [w(y) + f(y)] dy.$$

Using the asymptotic form of $G(x, y)$ as $|x| \rightarrow \infty$:

$$G(x, y) \sim \frac{e^{i\kappa r}}{\sqrt{r}} \underbrace{\frac{\gamma}{2\kappa^2} e^{-i\kappa \hat{x} \cdot y}}_{\text{far-field term}}, \quad \gamma = \frac{e^{i\pi/4}}{\sqrt{8\pi\kappa}}$$

Then the far-field pattern becomes:

$$w^\infty(\hat{x}) = \frac{\kappa^2}{2} \int_D e^{-i\kappa \hat{x} \cdot y} (n - 1) [f(y) + w(y)] dy.$$

Hence, for a Herglotz incident field $f = Hg$,

$$(\textcolor{red}{F}g)(\hat{x}) = \frac{\kappa^2}{2} \int_D e^{-i\kappa \hat{x} \cdot y} (n - 1) [\textcolor{green}{H}g(y) + w(y)] dy.$$

Recognizing the operator structure:

$$\textcolor{red}{F} = \textcolor{green}{H}^* \textcolor{blue}{T} \textcolor{green}{H}, \quad \textcolor{blue}{T}f = \frac{\kappa^2}{2} (n - 1)(f + w).$$

Factorization Method Works by Factorization!

- ▶ **Herglotz wave operator:** $H : L^2(\mathbb{S}^1) \rightarrow L^2(D)$, $Hg = v_g|_D$
- ▶ Define $T : L^2(D) \rightarrow L^2(D)$ by

$$Tf = \frac{\kappa^2}{2}(n-1)(f + w(f)|_D)$$

where $(\Delta^2 - \kappa^4 n(x))w = \kappa^4(n-1)f$ in $\mathbb{R}^2 +$ (SRCs)

Theorem

$$\begin{array}{ccc}
 & F = H^* T H & \\
 L^2(\mathbb{S}^1) & \xrightarrow{H} & L^2(D) \\
 \downarrow F & & \downarrow T \\
 L^2(\mathbb{S}^1) & \xleftarrow{H^*} & L^2(D)
 \end{array}$$

Task: Relate the Range(H^*) to the known operator F .

Key Property of the Middle Operator T

Operator factorization:

$$F = H^* T H, \quad T f = \frac{\kappa^2}{2} (n-1) (f + w)$$

Define the imaginary parts:

$$\Im(F) := \frac{F - F^*}{2i}, \quad \Im(T) := \frac{T - T^*}{2i},$$

which are self-adjoint & positive.

$$\Im(F) = H^* \Im(T) H$$

We have: $\text{Range}((\Im(F))^{1/2}) = \text{Range}(H^*)$, which follows from the following result.

Theorem ((Ceja-Ayala, R., Harris, I. & Ozochiawaeze, G. (2025))

Let the scatterer $D = \text{supp}(n-1)$ be **absorbing**, i.e.,

$$\Im(n(x)) \geq \alpha > 0 \quad \text{a.e. } x \in D.$$

Then the middle operator $\Im(T)$ is **coercive** on $L^2(D)$, that is, $\exists \mu > 0$ so that

$$(\Im(T)f, f)_{L^2(D)} = \Im(Tf, f)_{L^2(D)} \geq \mu \|f\|_{L^2(D)}^2$$

for all $f \in L^2(D)$.

Key. Proof by contradiction, exploit biharmonic wave decomposition for w obtained from factoring $(\Delta^2 - \kappa^4)w = (\Delta + \kappa^2)(\Delta - \kappa^2)w$

Reconstruction of D by the FM

We have the positive definite, compact operator

$$\mathfrak{I}(F) := \frac{F - F^*}{2i} : L^2(\mathbb{S}^1) \rightarrow L^2(\mathbb{S}^1).$$

We also have

$$\text{Range}((\mathfrak{I}(F))^{1/2}) = \text{Range}(H^*)$$

so

$$z \in D \iff \phi_z \in \text{Range}((\mathfrak{I}(F))^{1/2})$$

The Factorization Method (FM)

Let the pair $(\lambda_j, \psi_j) \in \mathbb{R}^+ \times L^2(\mathbb{S}^1)$ be the orthonormal eigensystem of $\mathfrak{I}(F)$. Then by the Picard range criterion we have:

$$z \in D \iff W(z) = \left[\sum_{j=1}^{\infty} \frac{|(\phi_z, \psi_j)_{L^2(\mathbb{S}^1)}|^2}{\lambda_j} \right]^{-1} > 0.$$

Thus, we have complete characterization of penetrable scatterer D :

$$\chi_D(z) = \text{sgn}(W(z)) := \begin{cases} 1, & x \in D \\ 0, & x \notin D \end{cases}$$

Born Approximation for Weak Scatterers

Integral operator:

$$(Kf)(x) := \int_D G(x, y) (n(y) - 1) f(y) dy$$

$$\text{Assume } \|K\| < 1, \quad (|D| < 1).$$

L-S Type Equation then becomes:

$$u^s = Ku^{\text{in}} + Ku^{\text{sc}} \iff u^s = (I - K)^{-1} Ku^{\text{in}}$$

Born approximation:

$$u^{\text{sc}} \approx \underbrace{Ku^{\text{in}}}_{\text{first-order}}$$

Far-field (linearized):

$$\begin{aligned} u^\infty(\hat{x}, d) &\approx \frac{\kappa^2}{2} \int_D (n - 1) e^{-i\kappa\hat{x}\cdot y} u^{\text{in}}(y) dy \\ &= \frac{\kappa^2}{2} \int_D (n - 1) e^{-i\kappa\hat{x}\cdot y} e^{i\kappa d\cdot y} dy \\ &= \kappa^2/2 \cdot \widehat{(n - 1)}(\xi), \quad \xi := \kappa(d - \hat{x}) \end{aligned}$$

Key intuition: small/weak scatterer $\Rightarrow$ far-field $\approx$ Fourier transform of contrast.

Comparing Exact Far-Field & Born Approximation

Assume a small disk scatterer $D = B_\epsilon$, $0 < \epsilon \leq 1$, with constant material parameter $n = 1.0 + 1.5i$. The direct scattering problem reads:

$$\Delta^2 u^{\text{sc}} - \kappa^4 u^{\text{sc}} = 0 \quad \text{in } \mathbb{R}^2 \setminus \overline{B_\epsilon}, \quad \Delta^2 u - \kappa^4 u = 0 \quad \text{in } B_\epsilon,$$

with boundary (transmission) conditions at $r = \epsilon$:

$$u^{\text{sc}} - u = -u^{\text{in}}, \quad \partial_r u^{\text{sc}} - \partial_r u = -\partial_r u^{\text{in}}, \quad \Delta u^{\text{sc}} - \Delta u = -\Delta u^{\text{in}}, \quad \partial_r \Delta u^{\text{sc}} - \partial_r \Delta u = -\partial_r \Delta u^{\text{in}}.$$

Expand incident plane wave via Jacobi–Anger:

$$u^{\text{in}}(r, \theta) = \sum_{\ell=0}^{\infty} i^\ell J_\ell(\kappa r) e^{i\ell(\theta-\phi)}, \quad d = (\cos \phi, \sin \phi)$$

Separation of variables and the boundary conditions at $r = \epsilon$ give a linear system:

$$\mathbb{M} \mathbf{u} = \mathbf{f}, \quad \mathbf{u} = [a_\ell, b_\ell, c_\ell, d_\ell]^T, \quad \mathbf{f} = [-J_\ell(\kappa\epsilon), -\kappa J'_\ell(\kappa\epsilon), \kappa^2 J_\ell(\kappa\epsilon), \kappa^3 J'_\ell(\kappa\epsilon)]^T.$$

Once solved, the exact biharmonic far-field pattern is

$$u^\infty(\theta, \phi) = \frac{4}{i} \sum_{\ell=0}^{\infty} a_\ell e^{i\ell(\theta-\phi)}.$$

Computing the Matrix $\mathbb{M}$ for a Disk Scatterer

Fourier Series Ansatz for the Scattered Field.

We decompose u^{sc} and u into propagative $((\Delta + \kappa^2)$ and evanescent $(\Delta - \kappa^2)$ components:

$$u_{\text{H}}(r, \theta) = \sum_{|\ell|=0}^{\infty} i^{\ell} a_{\ell} H_{\ell}^{(1)}(\kappa r) e^{i\ell(\theta-\phi)}, \quad u_{\text{M}}(r, \theta) = \sum_{|\ell|=0}^{\infty} i^{\ell} b_{\ell} H_{\ell}^{(1)}(i\kappa r) e^{i\ell(\theta-\phi)}.$$

Inside the Scatterer.

The total field $u = u_{\text{pr}} + u_{\text{ev}}$ is expanded as

$$u_{\text{pr}}(r, \theta) = \sum_{|\ell|=0}^{\infty} i^{\ell} c_{\ell} J_{\ell}(\kappa n^{1/4} r) e^{i\ell(\theta-\phi)}, \quad u_{\text{ev}}(r, \theta) = \sum_{|\ell|=0}^{\infty} i^{\ell} d_{\ell} J_{\ell}(i\kappa n^{1/4} r) e^{i\ell(\theta-\phi)}.$$

Key Observation.

$$\Delta u_{\text{pr}} = -\kappa^2 \sqrt{n} u_{\text{pr}}, \quad \Delta u_{\text{ev}} = \kappa^2 \sqrt{n} u_{\text{ev}}.$$

Linear System from Boundary Conditions.

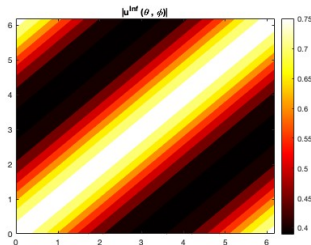
At $r = \epsilon$, the four boundary conditions give

$$\mathbb{M}\mathbf{u} = \mathbf{f},$$

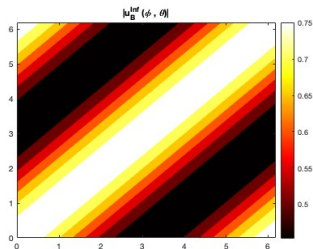
which determines the coefficients $a_{\ell}, b_{\ell}, c_{\ell}, d_{\ell}$.

Biharmonic Far-Field Pattern vs Born Approximation ($\epsilon = 0.5$)

Exact Far-Field Pattern



Born Approximation



$$\mathbf{F}_{\text{exact}} = [u^\infty(\theta_i, \phi_j)]_{i,j=1}^{64}, \quad \mathbf{F}_{\text{Born}} = [u_{\text{B}}^\infty(\theta_i, \phi_j)]_{i,j=1}^{64},$$

$$\hat{x} = (\cos \theta, \sin \theta), \quad d = (\cos \phi, \sin \phi), \quad n = 1.0 + 1.5i$$

Absolute Error $\|\mathbf{F}_{\text{exact}} - \mathbf{F}_{\text{Born}}\|_\infty$ for Small Disks B_ϵ

ϵ	Error
1.00	0.6480
0.90	0.5008
0.80	0.3784
0.70	0.2774
0.60	0.1935
0.50	0.2134

FM Discretization and Imaging Functional

Algorithm 1 Factorization Method

Input: Wavenumber κ , will fix regularization $\alpha = 10^{-5}$, noise level δ .

Step 1: Sample Directions.

Compute 64 equally spaced angles $\theta_i = 2\pi(i-1)/64$, $\hat{x}_i = d_i = (\cos \theta_i, \sin \theta_i)$.

Step 2: Assemble Far-Field Matrix.

Evaluate $\mathbf{F}(i, j) = u^\infty(\hat{x}_i, d_j)$ and form $\mathbf{F} \in \mathbb{C}^{64 \times 64}$.

Step 3: Add Noise (optional).

$\mathbf{F}^\delta(i, j) = \mathbf{F}(i, j)(1 + \delta \mathbf{R}(i, j))$, $\mathbf{R} \in \mathbb{C}^{64 \times 64}$ (error matrix)

Step 4: Compute SVD of Imaginary Part.

$\Im(\mathbf{F}) = \frac{\mathbf{F} - \mathbf{F}^*}{2i} = \sum_j \sigma_j \mathbf{u}_j \mathbf{v}_j^*$.

Step 5: FM Indicator.

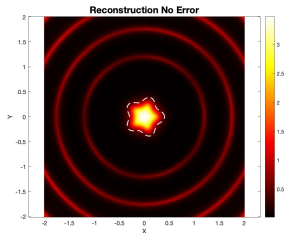
$$W_{\text{FM}}(z) = \left[\sum_j \frac{\phi^2(\sigma_j; \alpha)}{\sigma_j} |(\mathbf{u}_j, \ell_z)|^2 \right]^{-1}, \quad \ell_z = [e^{-i\kappa \hat{x}_i \cdot z}].$$

Tikhonov Filter Function

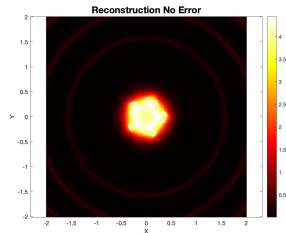
Tikhonov regularization: $\phi(t; \alpha) = \frac{t^2}{t^2 + \alpha}, \quad \alpha = 10^{-5}$

FM Reconstruction: Star-shaped Scatterer (Penetrable, $n = 2.5 + 1.5i$)

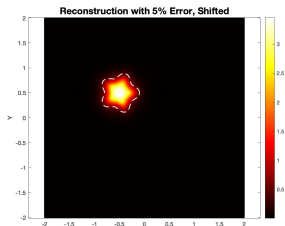
$k = 2\pi$ (no noise)



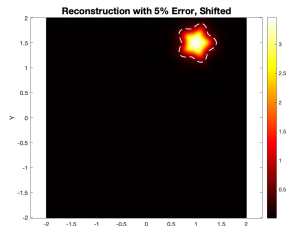
$k = 3\pi$ (no noise)



(5% noise, shifted (-0.5, 0.5))

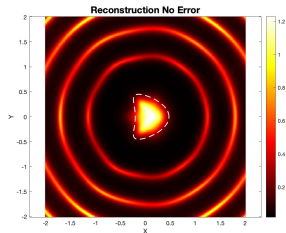


(5% noise, shifted (1, 1.5))

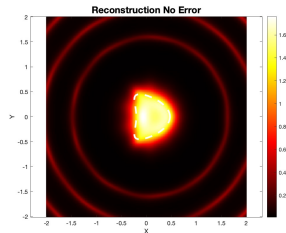


FM Reconstruction: Kite-shaped Scatterer (Penetrable, $n = 2.5 + 1.5i$)

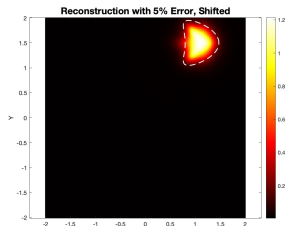
$k = 2\pi$ (no noise)



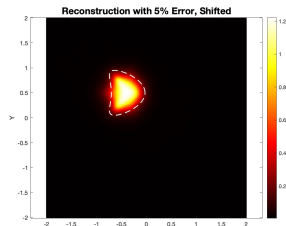
$k = 3\pi$ (no noise)



5% noise, shifted $(-0.5, 0.5)$

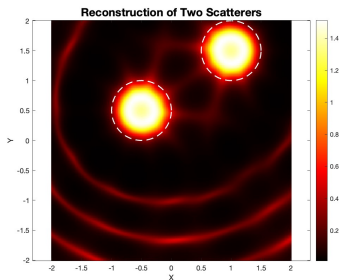


5% noise, shifted $(1, 1.5)$

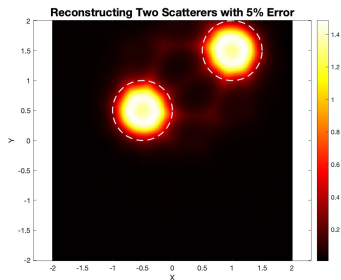


FM Reconstructions: Disk Scatterers (Penetrable, $\kappa = 2\pi$, $n = 2.5 + 1.5i$)

Reconstruction (No Noise)



Reconstruction (5% Noise)



FM reconstructions of a small disk scatterer $D = B_{1/2}$.

Conclusion

- ▶ Extended the **Factorization Method (FM)** to **biharmonic scattering** in **penetrable, absorbing media**.
- ▶ Provided **numerical reconstructions** showing FM's effectiveness for small scatterers and limited/noisy data.
- ▶ verified heuristically Born approximation valid first order linearization for weak scatterers.
- ▶ Open problems / future directions:
 - FM for **biharmonic scattering by impenetrable cavities** is still open.
 - Extending reconstruction methods to **near-field biharmonic scattering** for a penetrable medium remains open.
- ▶ Overall, FM provides a **qualitative reconstruction framework** beyond classical acoustic, elastic, & electromagnetic scattering, with extensions to complex media feasible.

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